CS110 (Principles of CS)
Recitation 5
Python Intro

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Section – 3 & 4

Agenda

1. **Warm-up & recap** Base conversions, setup, operators, interpreter
2. **Variables** How Python stores values
3. **Arithmetic & output** Basic operations and descriptive printing
4. **Libraries** Using random and math
5. **Boolean logic** True / False values
6. **Conditionals** Making decisions with if-else
7. **Assignment 1 prep** Rectangle problem, practice, and assignment review

Warm-up: Converting Between Bases

Decimal to Binary (divide by 2)

Divide	Quotient	Remainder
45 ÷ 2	22	**1**
22 ÷ 2	11	**0**
11 ÷ 2	5	**1**
5 ÷ 2	2	**1**
2 ÷ 2	1	**0**
1 ÷ 2	0	**1**

Read remainders bottom → top: **$45_{10} = 101101_2$**

Binary to Hex (groups of 4 bits)



101101_2 = **$2D_{16}$**

Check: 2 × 16 + 13 = 45 ✓

Pad on the left with 0s so every group has 4 bits.

Your Turn: Quick Conversions

1

Convert 13_{10} to binary

2

Convert 11110000_{2} to hex

3

Convert $1F_{16}$ to decimal

- Work with a neighbor for 3 minutes, then we check together
  - Hint: hex digits A–F stand for 10–15

Quick Recap: Getting Set Up

1. Python_programs folder

- On your Desktop: every program you write goes here

2. Pick your IDE

- **Python Playground:** `programiz.pro/ide/python`
- **VS Code (recommended):** office hours for setup help

3. Hello World

- Run it, download it (.zip), unzip, rename to Hello.py
- Upload Hello.py back to the IDE to test

4. Runestone textbook

- Course name: **`RU_thinkcspy_Fall24`**
- Bookmark the textbook URL

Numeric Operators

Operator	Meaning	Example	Result
`+`	Addition	`7 + 2`	**`9`**
`-`	Subtraction	`7 - 2`	**`5`**
`*`	Multiplication	`7 * 2`	**`14`**
`/`	Division	`7 / 2`	**`3.5`**
`**`	Exponent (power)	`7 ** 2`	**`49`**

Order of operations

- Same as algebra
- `**` first
- `*` `/` next
- `+` `-` last
- Parentheses change the order

Compare

```
2 + 3 * 4     →  14
(2 + 3) * 4 →  20
```

In VS Code: The Shell & Interpreter

- **Running a program** executes the whole .py file
- **The interpreter** answers one expression at a time
- **Try it:**
  - Open the terminal and split the window
  - Type **`python3`** (Windows: python) to enter immediate mode
  - Type a few arithmetic expressions
  - Type **`exit()`** to leave

Terminal

```
$ python3
>>> 2 + 3 * 4
14
>>> (2 + 3) * 4
20
>>> 2 ** 10
1024
>>> 7 / 2
3.5
```

Note: Python Playground only runs main.py and cannot do immediate mode.

Required Resources

Interactive textbook

- Sections 1.1 – 1.5
- Sections 2.1, 2.2
- Sections 2.4 – 2.7
- Section 2.9
- **Practice: 2.13 exercises**

Python Reference Guide

- `math` library: ceil, floor, sqrt
- `random` library: randint, random
- Keep it open while you code

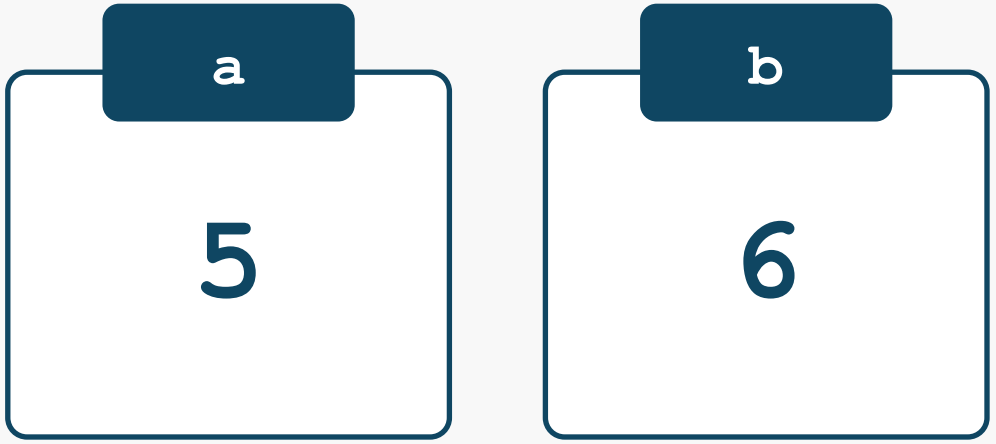
Canvas

- Python Study Guide starts in the Week 6 Module
- Study Guides / Resources for Python

Exam tip: doing ALL the assigned interactive exercises in the textbook is great review!

What Is a Variable?

- **A variable stores data**
  - Think of it as a labeled box that holds a value
- The = sign means "store", not "equals"
  - Right side is calculated first, then saved on the left
- Names should describe the value: length, width, area



Python

```
a = 5
b = 6
```

Example 1: Basic Addition

Code

```
a = 5
b = 6
c = a + b
print("The sum is:", c)
```

Output

```
The sum is: 11
```

Step by step

1. Values assigned to variables
2. Python adds the numbers
3. Result saved in a new variable
4. Output printed

Concepts: **Assignment** **Arithmetic** **Output**

Strings and Concatenation

Combine text and variable values to make descriptive output

Option 1: commas

```
c = 11
print("The sum is:", c)
```

Output

```
The sum is: 11
```

Option 2: + with str()

```
c = 11
print("The sum is: " + str(c))
```

Output

```
The sum is: 11
```

- **Commas** add a space automatically and work with any type
- **+** only joins text to text, so wrap numbers in **`str()`**

Importing Libraries

Libraries give Python extra tools. Import them at the top of your program.

At the top of your file

```
import random
import math
```

random: generate numbers

- `random.randint(1, 10)` → whole number 1–10 (both included)
- `random.random()` → decimal from 0 up to (not incl.) 1

math: math operations

- `math.ceil(x)` → round UP
- `math.floor(x)` → round DOWN
- `math.sqrt(x)` → square root

Example 2: Random Integers & Booleans

Code

```
import random
a = random.randint(1, 10)
b = random.randint(1, 10)
c = a + b
big = (c >= 10)      # Boolean value

print("numbers are:", a, b)
print("sum is big:", big)
```

Possible output (changes every run)

```
numbers are: 3 8
sum is big: True
```

Boolean values

- A logical result: True or False
- `11 >= 10` → True
- `7 >= 10` → False

Try it

Run it 3 times. Do the numbers change? Does big change?

Conditionals: Programs Make Decisions

Basic structure

```
if a > 5:
    print("big")
else:
    print("small")
```

Boolean variable with condition

```
bigNumber = a > 5    # True/False

if bigNumber:
    print("big")
else:
    print("small")
```

Why Booleans matter

- They let programs decide what to do
- Used in conditionals, decisions, and assignment problems

Rules to remember

- **Colon** after if / else, **indent** the body
- == compares, = stores

Example 3: Random Decimals, round & ceil

Code

```
import random
import math

a = random.random()
print(a)
print(round(a, 2))

b = a + random.randint(1, 4)
print(b)
print(math.ceil(b))
print(math.floor(b))
```

Possible output

```
0.347829
0.35
3.347829
4
3
```

Value	round(x, 2)	ceil()	floor()
0.3478	0.35	1	0
3.2	3.2	4	3
5.9	5.9	6	5

What to notice

- `random.random()` gives 0 ≤ value < 1
- Decimal + integer = decimal: Python handles mixed types
- round keeps decimals; ceil/floor give whole numbers

Sample Problem: A Random Rectangle

Steps 1–2: random integer sides

```
import random

length = random.randint(1, 10)
width = random.randint(1, 10)

area = length * width
perimeter = 2 * (length + width)

print("Length:", length, "Width:", width)
print("Area of the rectangle is:", area)
print("Perimeter of the rectangle is:", perimeter)

# discussion: expression directly
print("Area (direct):", length * width)
```

Sample output

```
Length: 4 Width: 7
Area of the rectangle is: 28
Perimeter of the rectangle is: 22
Area (direct): 28
```

Steps

- Generate random length and width
- Calculate area and perimeter
- Print descriptive output
- **Discuss:** store in a variable, or print length * width directly?
  - Variables can be reused and are easier to read

Step 3: Decimal Sides, ceil, floor & round

rectangle.py

```
import random
import math

# decimal numbers in [1, 10)
length = random.random() * 9 + 1
width = random.random() * 9 + 1
area = length * width

print("Length:", round(length, 2))
print("Width:", round(width, 2))
print("Area:", round(area, 2))
print("Ceiling:", math.ceil(area))
print("Floor:", math.floor(area))
```

Sample output

```
Length: 3.47
Width: 6.21
Area: 21.54
Ceiling: 22
Floor: 21
```

Why * 9 + 1?

- `random.random()` gives [0, 1)
- × 9 → [0, 9), + 1 → [1, 10)

Steps 4 & 5: Decide Big or Small

Simple conditional

```
if length > 5:
    print("big")
else:
    print("small")
```

With a Boolean variable

```
bigNumber = length > 5   # True/False
print("Is it big?", bigNumber)

if bigNumber:   # same as == True
    print("big")
else:
    print("small")
```

Remember

- **Colon** at the end of if / else
- **Indent** the code inside (4 spaces)
- **Quotes** around text to print

Boolean = True or False

A comparison like length > 5 produces a Boolean you can store, print, and test.

Spot the Bug!

Buggy code	Why it breaks	Fixed
`if bigNumber = True:`	= assigns, == compares	`if bigNumber == True:`
`print(big)`	Text needs quotes	`print("big")`
`print(“Hello”)`	Curly quotes from Word/PDF	`print("Hello")`
`"Area: " + area`	Can’t add text and a number	`"Area: " + str(area)`

Read the error message: it tells you the line number and the kind of error.

Before Assignment 1

Focus: expressions and assignment statements

- **Before you code, ask:**
  - What values do I need? (these become variables)
  - What do I calculate? (expressions)
  - What should the output look like? (descriptive print statements)
- **Start with the comment header:** name, section, honor code
- **Save your .py file in Python_programs**

Starter layout

```
# Name: ...
# Section: ...
# Honor code: ...

# 1. store values
# 2. calculate
# 3. print results
```

Practice Problems

1. Circle

- Random radius from 1 to 10
- Print area and circumference using `math.pi`
- Round to 2 decimals

2. Dice Roll

- Roll two dice (1 to 6 each)
- Print both dice and the sum
- Boolean `doubles`: print "Doubles!" or "No doubles"

3. Temperature

- Random °F from 0 to 100
- Convert to Celsius:
  - `C = (F-32)*5/9`
- Round to 1 decimal
- Print "Hot!" if F ≥ 80, else "Not hot"

Work in pairs: pick one problem, save it in Python_programs, and run it a few times to see different random values.

Thank You!

Questions?